AI for Learning and Teaching the Arabic Language:

Analytical Perspectives



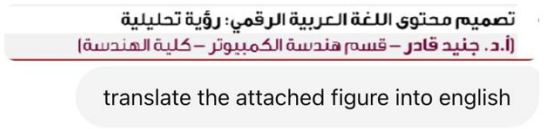
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JUNAID QADIR

Professor of Computer Engineering, Qatar University

AI is not a monolith: looking at the opportunities

Generative AI/ LLMs (popularized by ChatGPT)

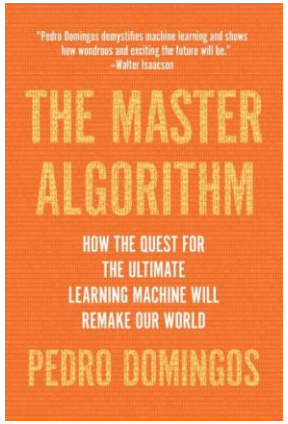


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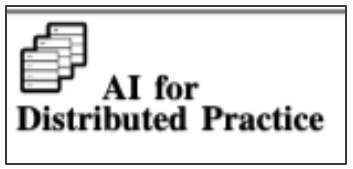
"Designing Digital Arabic Language Content: An Analytical Perspective
(Prof. Junaid Qadir – Department of Computer Engineering – College of Engineering)"

duolingo

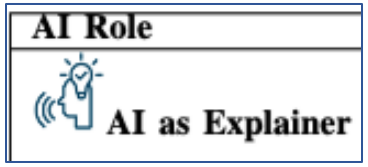
- PERSONALIZED LEARNING
- AI-POWERED SPEAKING PRACTICE
- SMART REVIEWS
- AUTO-EXERCISE GENERATION
- EXPLAIN MY ANSWER
- GAMIFICATION AND MOTIVATION
- A/B TESTING AND OPTIMIZATION
- AI AS A VIRTUAL TUTOR

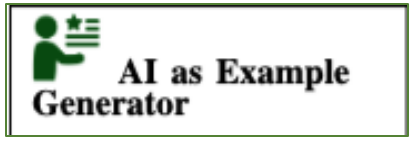


At least 5 Tribes:
—Symbolists
—Bayesians
—Connectionists
—Evolutionaries
—Analogizers

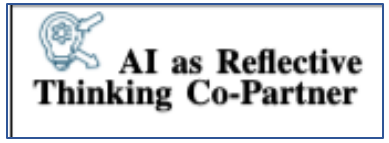












Language is also not a monolith: how GenAI can help

Integration of Traditional Language Skills (listening, reading, writing, and speaking) with ***Common European Framework of Reference for Languages (CEFR) Language Modes*** *(production, reception, interaction, and mediation)*

	Oral	Written
Receptive (Reception)	*Listening*	*Reading*
Productive (Production)	*Speaking*	*Writing*
Interaction	*Speaking and Listening*	*Reading and Writing*
Mediation	*Translating and Interpreting*	*Summarizing and Adapting*

CEFR Level	Proficiency Stage	GenAI Applications
A1	Basic User - Beginner	**Vocabulary Building**: Generate simple words and phrases in context.
		Basic Grammar: Provides structured grammar practice.
A2	Basic User - Elementary	**Simple Conversations**: Simulates everyday interactions (e.g., greetings, self-introduction).
		Listening Practice: Generates audio outputs for basic listening comprehension.
B1	Independent User - Intermediate	**Interactive Scenarios**: Engages learners in real-life scenarios (e.g., asking for directions).
		Expanded Vocabulary: Introduces topic-specific vocab for wider conversation topics.
B2	Independent User - Upper Intermediate	**Debate and Discussion Practice**: Enables discussions on familiar and abstract topics.
		Pronunciation Feedback: Uses voice recognition to refine pronunciation.
C1	Proficient User - Advanced	**Advanced Composition**: Assists with complex text generation (e.g., essays, reports).
		Complex Conversations: Facilitates nuanced discussions on abstract topics.
C2	Proficient User - Mastery	**Specialized Language**: Generates field-specific terminology (e.g., technical, academic).
		Cultural Nuances: Introduces idiomatic expressions and cultural context.



Using technology in education

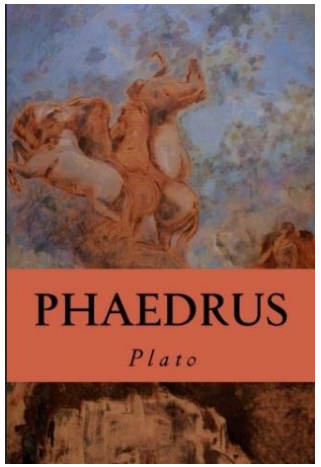
When it comes to technology in education, there are two camps:

(1) technological optimism ("this will solve everything") and

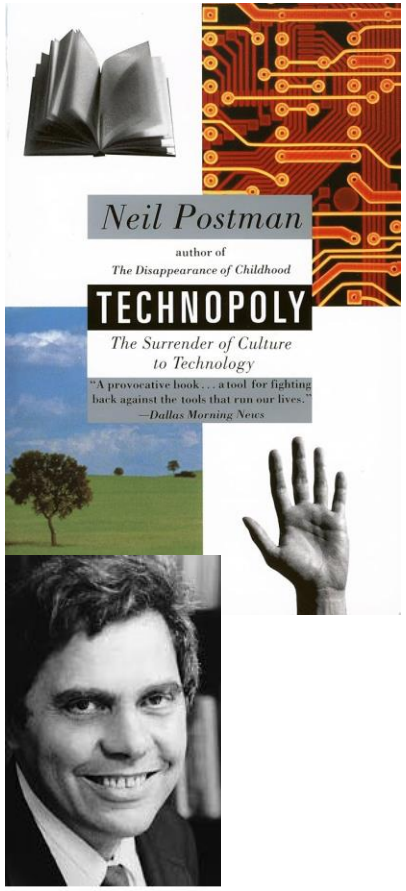
(2) ***technological skepticism*** *("this will ruin everything").*



More than 2,000 years ago, Plato dramatized this tension in his dialogue Phaedrus, through the myth of **Theuth — the inventor of writing — *and Thamus, the skeptical king*.**



Myth of Thamus and Theuth



🎓 Theuth —Inventor & Tech Enthusiast

Theuth: "This invention [writing], O King, will make the Egyptians wiser and will improve their memories; for it is an elixir of memory and wisdom."

🏛 Thamus – King, Technology Skeptic

"Theuth, my paragon of inventors, the discoverer of an art is not the best judge of the good or harm which will accrue to those who practice it. [...] Your invention will produce forgetfulness in the minds of those who learn to use it, because they will not practice their memory. [...] They will be hearers of many things and will have learned nothing; they will appear to be omniscient and will generally know nothing; they will be tiresome company, having the show of wisdom without the reality."

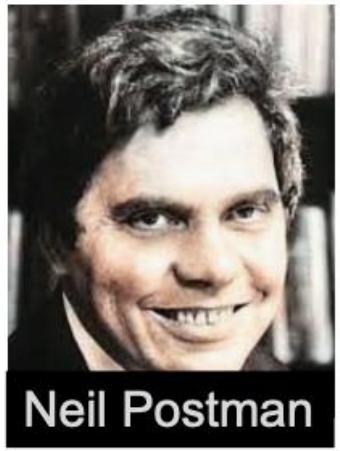
both of these protagonists had an important insight; but they were wrong in thinking that technology only works in a single way (either-or approach)

Two important insights about technology


Melvin Kranzberg

Kranzberg's First Law of Technology

"Technology is neither good nor bad;
nor is it neutral."
"we shape our tools and thereafter our tools shape us."


Neil Postman

"Every technology is
both a burden and a blessing;
not either-or, but this-and-that."

How to (and not to) navigate complexity ..

Procrustean Fallacy

Stretching or Constricting
Reality to Fit the Model



When dealing with a complex phenomena such as education and human learning, it is wise to adopt a "**both-and**" *multiplex mindset*, instead of a Procustrean reductionist "**either-or**" *uniplex mindset*.

Two vignettes on technology enhanced learning

AL-GHAZALI AND THE ROBBERS



Al-Ghazali stopped relying on books. But later on, Al-Ghazali used writing technology effectively to disseminate his knowledge.

PRINTING PRESS AND THE MUSLIM EMPIRE



Can we learn something from that experience for how we should deal with GenAI technology?

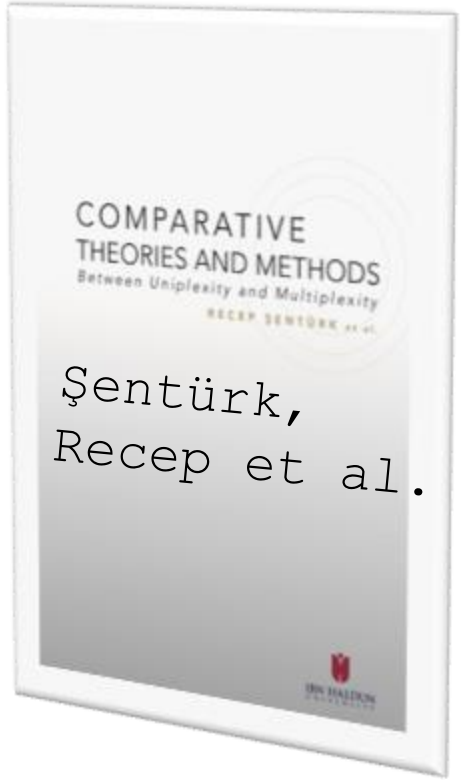
Does the technology strengthen or weaken our awareness, experience, and relationship with ourselves, our purpose, others, the world?

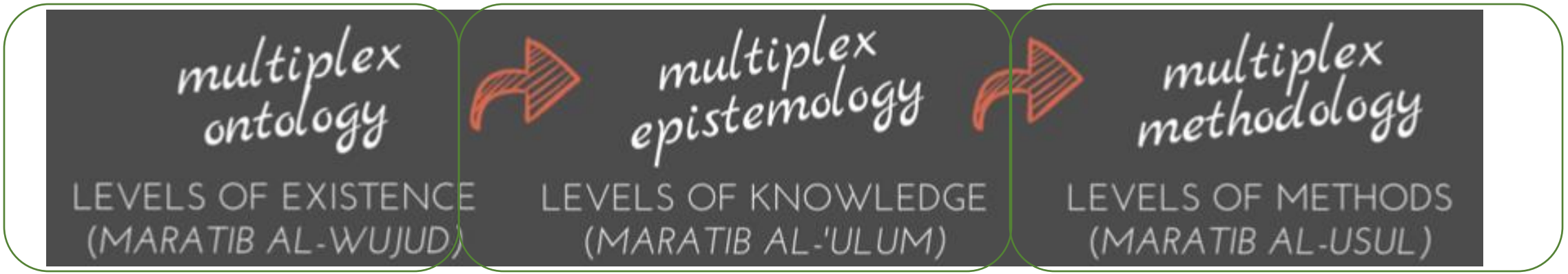
Argument for Multiplex Both-And Thinking

Multiplexity is a translation of the Arabic term "marātib", which literally means hierarchy or levels.

Multiplexity provides a multi-layered understanding of reality, knowledge, and truth, and language.

*Supports **Both-And Thinking** instead of Either-or Thinking.*





Principles for AI-Augmented Language Learning



Principle 1: AI is Both a Blessing and A Burden: Protect and Promote Human Creativity

AI should inspire originality—not standardize student voice.

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Use AI in service of what matters most: connection, understanding, purpose, meaning, wisdom.

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Thanks! jqadir@qu.edu.qa